



FIGURE 5.25 Tablas, or yarn paintings, created by members of the Huichol tribe (Mexico), evoke the beautiful lights, vivid colors, and “peculiar creatures” experienced after the people eat the hallucinogenic peyote cactus in highly ritualized ceremonies.

intense mystical experience. Peyote, a hallucinogen derived from a cactus plant, has a long history of use in Native American religious ceremonies.

LSD attaches mainly to one kind of brain receptor sensitive to the neurotransmitter serotonin (Jacobs, 1987). It stimulates those receptors at irregular times and prevents the brain's neurotransmitters from stimulating the receptors at the normal times. We have an interesting gap in our knowledge at this point: We know that LSD's disruption of certain kinds of serotonin receptors leads to hallucinogenic experiences, but we do not understand the relationship between those receptors and the resulting experiences. That is, we know where the hallucinogenic event happens, but not *how* it happens.

Chronic Drug Effects

The intended effects of any drug are temporary. Initially, they produce an effect, such as excitement, relaxation, or a distortion of experience. As the drug leaves the brain and the effects wear off, the person experiences **withdrawal effects**, which are generally *about the opposite of the initial effects*. After someone has taken a drug repeatedly, its effects grow weaker and weaker, unless the person increases the dosage. This *decrease in effect* is called **tolerance**. Drug users often seek the drug partly to combat the withdrawal effects; they increase their dosage to compensate for their tolerance. That is, withdrawal effects and tolerance tend to promote increased use of a drug.

Drug Withdrawal

When habitual users suddenly stop using alcohol or opiate drugs (such as morphine or heroin), they gradually enter a state of withdrawal (Gawin & Kleber, 1986). With alcohol, the typical withdrawal symptoms are sweating, nausea, sleeplessness, and, in severe cases, hallucinations and seizures (Mello & Mendelson, 1978). With opiate drugs, the typical withdrawal symptoms are anxiety, restlessness, loss of appetite, vomiting, diarrhea, sweating, and gagging (Mansky, 1978). People who quit using tranquilizers can experience sleeplessness and nervousness. Users who quit cocaine or other stimulant drugs experience a state of depression. Someone who *feels compelled to use a drug to reduce unpleasant withdrawal symptoms* is said to have a **physical dependence** on the drug.

We distinguish between physical dependence and **psychological dependence**, which is a *strong repetitive desire for something without any accompanying physical symptoms of withdrawal*. For example, habitual gamblers have a psychological dependence on placing bets, even though they can abstain from gambling without undergoing anything like the effects of “cold turkey” heroin withdrawal. A psychological dependence can be extremely insistent, and in many cases, it is pointless to try to decide whether someone's dependence is physical or psychological. (For example, the withdrawal effects for someone who quits using cocaine include mostly “psychological” effects such as a lack of pleasure, with only fairly brief physiological effects. So is the dependence physiological or psychological?)

Drug Tolerance

People who take a drug repeatedly develop a tolerance to its effects. To achieve the desired high, drug users must steadily increase the dose. Some longtime users inject three or four times more heroin or morphine into their veins than it would take to kill a nonuser.

What brings about drug tolerance? It may result in part from automatic chemical changes that occur in cells throughout the body, to counteract the drug's effects (Baker & Tiffany, 1985). It may also result in part from psychological causes. For example, alcohol impairs the coordination of rats as well as that of humans. If rats are simply injected with alcohol every day for 24 days, the rats show no apparent tolerance for alcohol, as indicated by a coordination test given on the last day. However, if their coordination is tested after each of the 24 injections, each test session offers the rats an opportunity to practice their coordination, so their performance steadily improves (Wenger, Tiffany, Bombardier, Nicholls, & Woods, 1981). In other words, by practicing coordination while under the influence of alcohol, the rats develop a tolerance to alcohol. Similarly, people who use stimulant drugs as appetite suppressants start to develop a tolerance to the effects after a few meals. We shall discuss drug tolerance in more detail in the next chapter.

CONCEPT CHECK

6. People who use stimulant drugs as appetite suppressants ordinarily develop a tolerance. How could they prevent the development of this tolerance? (Check your answer later on this page.)

THE MESSAGE Drugs and Awareness

If you were to change a few of a computer's connections at random, you could produce an "altered state," which would almost certainly not be an improvement. Giving drugs to a human brain is a little like changing the connections of a computer, and almost any drug at least temporarily impairs brain functioning somehow, even if the drug is used under medical supervision and accomplishes some good along with the bad. By examining both the desirable and undesirable effects of drugs on the brain, we can gain greater insight into the brain's normal processes and functions.

SUMMARY

- ★ **Alcohol.** Alcohol, the most widely abused drug in our society, relaxes people and relieves their inhibitions. It can also impair judgment and reasoning. (page 182)
- ★ **Tranquilizers.** Benzodiazepine tranquilizers are widely used to relieve anxiety; they are sometimes also used to relax muscles or to promote sleep. (page 183)
- ★ **Opiates.** Opiate drugs bind to endorphin receptors in the nervous system. The immediate effect of opiates is pleasure and relief from pain. (page 183)
- ★ **Marijuana.** Marijuana's active compound, THC, acts on abundant receptors, found mostly in the hippocampus and certain brain areas important for the control of movement. Because the medulla has few THC receptors, a large dose of marijuana is seldom fatal. (page 183)
- ★ **Stimulants.** Stimulant drugs such as amphetamines and cocaine increase activity levels and pleasure. Compared to other forms of cocaine, crack produces more rapid effects on behavior, greater risk of addiction, and greater risk of damage to the heart and other organs. (page 184)
- ★ **Hallucinogens.** Hallucinogens induce sensory distortions. LSD acts at one type of serotonin synapse; we do not yet know why activity at that type of synapse should produce these effects. (page 185)
- ★ **Withdrawal.** After using a drug, the user enters a rebound state known as withdrawal. Drug users often crave drugs as a way of decreasing their withdrawal symptoms. (page 186)
- ★ **Tolerance.** People who use certain drugs repeatedly become less and less sensitive to them over time. (page 186)

Suggestion for Further Reading

Rivers, P. C. (1994). *Alcohol and human behavior*. Englewood Cliffs, NJ: Prentice-Hall. Covers all aspects of alcohol use from its effects on physiology to its role in society and culture.

Terms

- alcohol** a class of molecules that includes methanol, ethanol, propyl alcohol (rubbing alcohol), and others (page 182)
- tranquilizers** drugs that help people to relax (page 183)
- opiates** either drugs derived from the opium poppy or synthetic drugs that produce effects similar to those of opium derivatives (page 183)
- endorphins** chemicals produced by the brain that have effects resembling those of opiates (page 183)
- stimulants** drugs that boost energy, heighten alertness, increase activity, and produce a pleasant feeling (page 184)
- hallucinogens** drugs that induce sensory distortions (page 185)
- withdrawal effects** experiences that occur as a result of the removal of a drug from the brain (page 186)
- tolerance** the weakened effect of a drug after repeated use (page 186)
- physical dependence** a condition whereby a habitual drug user is driven to seek the drug to escape or avoid the unpleasant withdrawal effects that occur during abstinence from the drug (page 186)
- psychological dependence** a strong repetitive desire for something without any physical symptoms of withdrawal (page 186)

Answer to Concept Check

6. Instead of taking a pill just before a meal, people should take it between meals, when they are not planning to eat right away or when they plan to skip a meal altogether. If they eat right after taking a pill, they soon develop a tolerance to its appetite-suppressing effects. However, even if they follow the advice to take the pills between meals, most people are not likely to lose weight in the long run. After their appetite is suppressed for a while, they are likely to experience increased appetite later. (page 187).

Web Resources

Web of Addictions

www.well.com/user/woa/

Andrew L. Homer and Dick Dillon provide factual information about alcohol and other abused drugs. Fact sheets and other material are arranged by drug, with links to net resources related to addictions, in-depth information on special topics, and a list of places to get help with addictions.

National Clearinghouse for Alcohol and Drug Information

www.health.org/survey.htm

Dozens of research and statistical reports about alcohol and drug use. Backtrack to Prevention Online's front page to access even more information, searchable databases, forums, and links.

Learning



6

★ MODULE 6.1

Behaviorism

The Rise of Behaviorism

The Assumptions of Behaviorism

Determinism / The Ineffectiveness of Mental Explanations / The Power of the Environment to Mold Behavior

★ MODULE 6.2

Classical Conditioning

Pavlov and Classical Conditioning

Pavlov's Procedures / The Phenomena of Classical Conditioning / Stimulus Generalization

Drug Tolerance as an Example of Classical Conditioning

Explanations of Classical Conditioning

Pavlov's Theory of the Causes of Classical Conditioning / A Signal, Not Just a Transfer of Responses / CONTIGUITY ALONE OR CONTINGENCY: WHAT'S THE EVIDENCE? / Conditioning, Contiguity, and Contingency

★ MODULE 6.3

Operant Conditioning

Thorndike and Operant Conditioning

Extinction, Generalization, Discrimination, and Discriminative Stimuli / Why Are Certain Responses Learned More Easily Than Others?

B. F. Skinner and the Shaping of Responses

Shaping Behavior / Chaining Behavior

Increasing and Decreasing the Frequency of Responses

Reinforcement and Punishment / What Constitutes Reinforcement? / Schedules of Reinforcement

Practical Applications of Operant Conditioning

Animal Training / Persuasion / Applied Behavior Analysis / Behavior Modification / Breaking Bad Habits

★ MODULE 6.4

Other Kinds of Learning

Conditioned Taste Aversions

Birdsong Learning

Social Learning

Modeling and Imitation / Vicarious Reinforcement and Punishment / The Role of Self-Efficacy in Social Learning / Self-Reinforcement and Self-Punishment in Social Learning

Suppose we set up a simple experiment on animal learning. We put a monkey midway between a green wall and a red wall. If it approaches the green wall, we give it a few raisins; if it approaches the red wall, it gets nothing. After a few trials, the monkey always approaches the green wall. After it has made the correct choice, say, 12 times in a row, we are satisfied that the monkey has learned something.

Now let's suppose that we conduct the same experiment with an alligator. We use the same procedure as we did with the monkey, but we get different results. The alligator strains our patience, sitting for hours at a time without approaching either wall. When it finally moves, it is as likely to approach one wall as the other. After hundreds of trials, we have learned something: not to go into the trained-alligator business. But we see little evidence that the alligator has learned anything.

Does this mean that alligators are slow learners? Not necessarily. Maybe they are just not motivated to seek food. Maybe they can't see the difference between red and green. Maybe they learn but also forget quickly, so they cannot put

together a long streak of consecutive correct approaches. To decide for sure whether alligators can learn, we would need to test them under a wide variety of circumstances. (I don't know whether you care, but yes, alligators can learn. They're not much good at learning to find food, but according to Davidson [1966], they *can* learn how to get away from unpleasant stimuli.)

Some similar problems arise when evaluating human learning. Suppose that little Joey is having serious academic troubles. Should we consider him a "slow learner"? Not necessarily. Like the alligator, Joey may not be sufficiently motivated, or he may have trouble seeing or hearing, or he may learn and then forget. Maybe he is distracted by his emotional troubles at home. (That's a possibility we probably wouldn't consider with the alligator.) We can imagine all sorts of other reasons why Joey might be having trouble in school.

Psychologists have spent an enormous amount of time studying learning. One of their main discoveries is how important it is to consider all the influences that might interfere with learned performance. Psychologists have developed and polished many of their skills by conducting experiments on learning.

This chapter is about the procedures that produce changes in behavior—why you lick your lips at the sight of tasty food, why you turn away from a food that once made you sick, why you get nervous if a police car starts to follow you, and why you shudder at the sight of a ferocious person with a chain saw. This chapter is also about why you work harder at some tasks than at others and why you sometimes persevere for so long before you give up. Chapter 7 deals with memories. Obviously, one cannot learn a change in behavior without establishing some sort of memory, and one cannot establish a memory without learning something. Still, the study of learning is based on a very different research tradition from that of memory.



Behaviorism

How and why did the behaviorist viewpoint arise?

What is its enduring message?

Different kinds of psychologists disagree about the primary goals of psychology. For example, in later chapters we shall encounter *humanistic psychologists*, who are mainly interested in people's personal experiences and values. Here, we discuss **behaviorists**, *psychologists who insist that psychologists should study only observable, measurable behaviors, not mental processes*. The discussion of behaviorism will lead us into a discussion of learning, a field of research that behaviorists have traditionally dominated.

Because the term *behaviorist* applies to theorists and researchers with quite a range of views, psychologists distinguish between *methodological behaviorists* and *radical behaviorists*. **Methodological behaviorists study only the events that they can measure and observe**—in other words, the environment and the individual's actions—but *they sometimes use those observations to make inferences about internal events* (Day & Moore, 1995). For example, depriving an animal of food, or increasing its exercise, or presenting very appealing food will increase the probability that the animal will eat food, work for food, and so forth. From such observations, a psychologist can infer an **intervening variable**, *something that we cannot directly observe but that links a variety of procedures to a variety of possible responses*. In this case, the intervening variable is *hunger*:

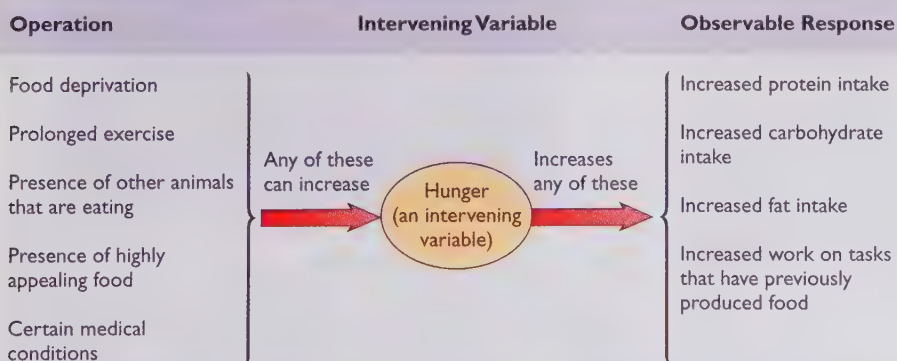
Similarly, one could use other kinds of observations to infer other intervening variables, such as thirst, sex drive, anger, fear, and so forth. The important point is that a methodological behaviorist will use such terms only after anchoring them firmly to observable procedures and responses—that is, after giving them a clear operational definition (as discussed in Chapter 2). Most psychological researchers are methodological behaviorists, even if they do not use that term.

Radical behaviorists do not deny that private events such as hunger, anger, or fear exist, and they also agree that it is possible to study the circumstances that cause people to say “I am hungry,” “I am angry,” and so forth. The distinguishing feature of radical behaviorists is that they *deny that hunger, anger, or any other internal, private event causes behavior* (Moore, 1995). For example, they would say, if food deprivation leads to hunger and hunger leads to eating, what have we gained by introducing the word *hunger*? Why not just say that food deprivation leads to increased eating? According to radical behaviorists, all internal states are caused by an event in the environment (or by the individual's genetics), and therefore the ultimate cause of any behavior lies in the observable events that led up to that behavior, not the internal states.

According to this point of view, most discussions of mental events are just sloppy language. For example, as B. F. Skinner (1990) argued, when you say, “I *intend* to . . .,” what you really mean is “I am about to . . .,” or “In situations like this, I usually . . .,” or “This is in the preliminary stages of happening. . . .” That is, any statement about intentions or mental experiences can be converted into a description of behavior.

The Rise of Behaviorism

Behaviorism can be clearly understood only within the historical context in which it arose. During the early 1900s, one highly influential group within psychology was the *structuralists* (see Chapter 1) who studied thoughts and ideas by asking people to describe their own





Behaviorists agree that all psychological investigations should be based on behavioral observations. A methodological behaviorist might use observations of, say, facial expressions to make inferences about such processes as “sadness.” A radical behaviorist, however, would study the facial expressions themselves, but not as a means of inferring something else.

experiences. The early behaviorists were, to a large extent, protesting against structuralism. Behaviorists insisted that it is useless to ask people to report on their own private experiences. For example, if someone says, “My idea of roundness is stronger than my idea of color,” we have no way to check the accuracy of the report. We are not even certain what it means. If psychology is to be a scientific enterprise, behaviorists insisted, it must deal with only observable, measurable events—that is, behaviors and the environment.

To avoid any mention of the mind, thoughts, or knowledge, some behaviorists went to the opposite extreme. One of the forerunners of behaviorism, Jacques Loeb (1918/1973), argued that much of animal behavior, and perhaps human behavior as well, could be described in terms of simple responses to simple stimuli—for example, approaching light, turning away from strong smells, clinging to hard surfaces, walking toward or away from moisture, and so forth. (See Figure 6.1.) Complex behavior, he surmised, is just the result of adding together many changes of speed and direction elicited by various stimuli. Loeb’s view of behavior was an example of **stimulus-response psychology**, *the attempt to explain behavior in terms of how each stimulus triggers a response*.

Although the term *stimulus-response psychology* was an appropriate term for Loeb, it is an inaccurate and misleading description of the behaviorists of today. Behaviorists do not believe that we can point to a simple stimulus to explain

every action. Behavior is a product of the individual’s past history of stimuli and responses, not simply a product of current stimuli.

If the behaviorist approach is to deal successfully with complex behaviors, it must be able to explain changes in behavior that depend on learning. The behaviorist movement, which was developing during the early 1900s, became the heir to a tradition of animal learning research that had begun for quite different reasons. Charles Darwin’s theory of evolution by natural selection inspired many early psychologists to study animal learning and intelligence (Kalat, 1983). At first, those psychologists were interested in comparing the intelligence of various species. By about 1930, however, most psychologists had lost interest in that topic, because the differences in learning performance among species proved to be small and inconsistent. Nevertheless, the early researchers had begun a tradition of studying animal learning in an objective manner; behaviorists carried forth this tradition, although they asked different questions. If nonhumans learn in more or less the same way as humans do, behaviorists reasoned, then it should be possible to discover the basic laws of learning by studying the behavior of a convenient laboratory animal, such as a pigeon or a rat. Most of the rest of this chapter will deal with behaviorists’ attempts to outline the basic laws of learning.



FIGURE 6.1 Jacques Loeb, an early student of animal behavior, argued that much or all of invertebrate behavior could be described as responses to simple stimuli, such as approaching light, turning away from light, or moving opposite to the direction of gravity.

The Assumptions of Behaviorism

Behaviorists make several assumptions, including the validity of the determinist position, the ineffectiveness of mental explanations, and the power of the environment to select behaviors (Moore, 1995). Let's consider each of these points:

Determinism

Behaviorists assume that every behavior has causes that can be understood through scientific methods. In other words, behavior is lawful. One example of a behavioral law is that an individual will increase the rate of a behavior that leads to food and decrease a behavior that leads to a shock. Behaviorists seek to increase their understanding of the laws of behavior, and they test their understanding by trying to predict or control behavior. However, as you will remember from the discussion of free will and determinism in Chapter 1, in complex situations no one can expect completely accurate predictions.

The Ineffectiveness of Mental Explanations

In everyday life, we commonly “explain” our behaviors in terms of their motivations, or emotions, or our mental state. However, behaviorists insist that such statements explain nothing:

Q. Why did she yell at that man?

A. She yelled because she was angry.

Q. How do you know she was angry?

A. We can tell she was angry because she was yelling.

Clearly, references to mental states risk luring us into circular reasoning instead of a more fruitful explanation. To understand the causes of behavior, we should look to the environment and the individual's history. To avoid mental explanations, behaviorists either avoid mental terms altogether or use them very cautiously. B. F. Skinner, the most famous and influential behaviorist, resisted using even apparently harmless words such as *hide* because they imply an intention (L. Smith, 1995). Skinner preferred simply to describe what the individuals *did*, instead of guessing what they were *trying* to do.

The same insistence on description is central to both the British and American legal systems: A witness is asked “What did you see and hear?” An acceptable answer would be, “The defendant was sweating and trembling, and his voice was wavering.” It would be unacceptable to say, “The defendant was nervous and worried,” because such a statement requires an inference that the witness is not entitled to make. (Of course, after the witness describes the defendant's behavior, the judge or jury might draw a conclusion

from it, but at least they know by then what evidence exists to support it.)

The Power of the Environment to Mold Behavior

Behaviors produce outcomes. Eating your carrots has one kind of outcome; insulting your roommate has another. Depending on the kind of outcome, a behavior will occur either more or less often in the future. In effect, our environment selects successful behaviors, much as evolution selects successful animals.

Behaviorists have sometimes been accused of believing that the environment controls practically all aspects of behavior. The most extreme statement of environmental determinism came from John B. Watson, one of the founders of behaviorism, who said, “Give me a dozen healthy infants, well-formed, and my own specified world to bring them up in and I'll guarantee to take any one at random and train him to become any type of specialist I might select—doctor, lawyer, artist, merchant-chief, and yes, even beggar-man thief, regardless of his talents, penchants, tendencies, abilities, vocations, and race of his ancestors” (Watson, 1925, p. 82).

Watson admitted that his statement was an exaggeration. He defended himself by saying that many other people had similarly exaggerated the role of heredity in molding behavior. Today, few psychologists would claim that variations in behavior depend entirely on the environment (or that they depend entirely on heredity, for that matter). Although behaviorists today do not deny the importance of heredity, however, they do not generally emphasize it. Their research focuses on how the environment selects one behavior over another, and their explanations of individual differences concentrate on how different people's behaviors emerge from different learning histories.



Behaviorists emphasize the role of experience in determining our actions—both our current experience and our past experiences in similar situations.

THE MESSAGE

Behaviorism as a Method and Viewpoint

Some behaviorists take more extreme positions than others do. The same could be said for liberals, conservatives, environmentalists, vegetarians, or advocates of any other theoretical position. Just as politicians often attack their opponents by exaggerating their most extreme positions, some psychologists dismiss behaviorism by attacking some of the extreme statements of John Watson, Jacques Loeb, or other early behaviorists. Others abandon behaviorism because they want to investigate knowledge, imagination, or other internal processes, and they find behaviorist procedures and terminology too limiting.

Many students quickly dismiss behaviorism also, because, at least at first glance, it seems so ridiculous: “What do you *mean*, my thoughts and beliefs and emotions don’t cause my behavior?!” The behaviorists’ reply is, “Exactly right. Your thoughts and other internal states do not cause your behavior, because events in your present and past environment caused those thoughts. The events that caused those thoughts are therefore the real causes of your behavior, and psychologists should spend their time trying to understand the effects of those events, not trying to analyze your thoughts.”

Don’t be too quick to agree or disagree. Just contemplate this: If you believe that your thoughts or other internal states cause behaviors and that these behaviors were *not* caused by previous events in the environment, what evidence could you provide to support your claim?

SUMMARY

★ *Range of positions among behaviorists.* Behaviorists insist that psychologists should study behaviors and their relation to observable features of the environment. Methodological behaviorists use these observables to draw inferences about internal states. Radical behaviorists insist that internal states are of little scientific use and that they do not control behavior. The causes of the internal states themselves, as well as of the behaviors, lie in the environment. (page 191)

★ *The origins of behaviorism.* Behaviorism began in part as a protest against structuralists, who were asking people to describe their own mental processes. Behaviorists insisted

that the structuralist approach was futile and that psychologists should study observable behaviors. (page 191)

★ *Behaviorists’ interest in learning.* Before the rise of the behaviorist movement, other psychologists had studied animal intelligence. Behaviorists adapted some of the methods used in previous studies but changed the questions, concentrating on the basic mechanisms of learning. (page 192)

★ *Behaviorists’ assumptions.* Behaviorists assume that all behaviors have causes (determinism), that mental explanations are unhelpful, and that the environment acts to select effective behaviors and to suppress ineffective ones. (page 193)

Suggestions for Further Reading

Skinner, B. F. (1974). *About behaviorism*. New York: Knopf.

Skinner’s elaboration of the behaviorist point of view.

Todd, J. T., & Morris, E. K. (Eds.) (1995). *Modern perspectives on B. F. Skinner and contemporary behaviorism*. Westport, CT: Greenwood Press. A collection of articles in which leading behaviorists assess the past and future of the behaviorist movement.

Terms

behaviorist a psychologist who insists that psychologists should study only observable, measurable behaviors, not mental processes (page 191)

methodological behaviorist a psychologist who studies only measurable, observable events but sometimes uses those observations to make inferences about internal events (page 191)

intervening variable something that we infer without directly observing it and that links a variety of procedures to a variety of possible responses (page 191)

radical behaviorist a behaviorist who denies that internal, private events are causes of behavior (page 191)

stimulus-response psychology a field that attempts to explain behavior in terms of how each stimulus triggers a response (page 192)

Web Resources

Operant Conditioning and Behaviorism

www.biozentrum.uni-wuerzburg.de/genetics/behavior/learning/behaviorism.html

This site provides a historical outline that begins with Thorndike’s trial-and-error learning and includes Pavlov’s classical conditioning and Skinner’s operant conditioning.

Classical Conditioning

When we learn a relationship between two stimuli, what happens?

Do we start responding to one stimulus as if it were the other?

Or do we learn how to use information from one stimulus to predict something about the other?

You are sitting in your room when your roommate flicks a switch on the stereo. You flinch because you know the volume is set to a deafening level. In a case like this, you are not just responding to the flick of the switch; you are responding to what it predicts.

Certain aspects of our behavior consist of learned responses to signals. However, even apparently simple responses to simple stimuli no longer seem as simple as they once did. To explain even the simplest learned responses, we must give an individual credit for having processed a great deal of information.

Psychologists' efforts to discover what takes place during learning have led them to conduct thousands of experiments, many of them on nonhuman animals. For certain kinds of

learning, such as birdsong learning, the results depend heavily on which species is being studied, but for many other kinds of learning, the similarities among species are more impressive than the differences. In some ways, it is easier to study rats or pigeons than humans, because a researcher can better control what and when they eat and many other variables likely to influence performance. Besides, to the extent that species do differ, the behavior of a rat or a pigeon will probably be easier to understand than that of a human. Of course, if researchers discover great complexities in the behavior of a rat, they can safely assume that human behavior is at least equally complex.

Pavlov and Classical Conditioning

Suppose you always feed your cat at 4:00 P.M. with food you keep in the refrigerator. As 4:00 P.M. approaches, your cat goes to the kitchen, claws at the refrigerator, meows, and salivates. You might explain the cat's behavior by saying that it "expects" food, that it "knows" there is food in the refrigerator, or that it is "trying to get someone to feed it." Behaviorists reject such mental explanations and instead seek descriptions that do not rely on intentions, thoughts, or knowledge.

When Ivan P. Pavlov proposed a simple, highly mechanical theory of learning, the mood of the time was ready for his message. Pavlov, a Russian physiologist, won a Nobel Prize in physiology in 1904 for his research on digestion. As he continued his research, one day he noticed that a dog would salivate or secrete stomach juices as soon as it saw the lab worker who ordinarily brought the dogs their food. Because this secretion presumably depended on the dog's previous experiences, Pavlov called it a "psychological" secretion. Puzzled, Pavlov enlisted the help of other specialists, who then discovered that "teasing" a dog with the sight



Ivan P. Pavlov (with the white beard) with students and an experimental dog. Pavlov focused on limited aspects of the dog's behavior—mostly salivation—and devised some apparently simple principles to describe that behavior.

of food produced the same quantity and quality of saliva as food in the mouth did and that the secretion was as predictable and automatic as any reflex. Pavlov adopted the term *conditional reflex*, thus implying that he only *conditionally* (or tentatively) accepted it as a real reflex (Todes, 1997). However, this term has usually been translated into English as *conditioned reflex*, a term with different connotations.

Pavlov's Procedures

Pavlov guessed that animals are born with certain *automatic connections*—we call them **unconditioned** (or *unconditional*) **reflexes**—*between a stimulus such as food and a response such as secreting digestive juices*. He conjectured that animals acquire new reflexes by transferring a response from one stimulus to another. For example, if a neutral stimulus—say, a buzzer—always preceded food, an animal might begin to respond to the buzzer just as it responds to food. Thus, the buzzer would also elicit digestive secretions. The *process by which an organism learns a new association between two paired stimuli—a neutral stimulus and one that already evokes a reflexive response*—has come to be known as **classical conditioning** or **Pavlovian conditioning**. (It is called classical because it has been known and studied for a long time.)

Pavlov used an experimental setup like the one in Figure 6.2 (Goodwin, 1991). First, he selected dogs with a moderate degree of arousal. (Highly excitable dogs would not hold still long enough, and highly inhibited dogs would fall asleep.) Then, he attached a tube to one of the salivary ducts in the dog's mouth, to measure salivation. He could have measured stomach secretions, but it was easier to measure salivation.



FIGURE 6.2 Pavlov used dogs for his experiments on classical conditioning and salivation. The experimenter can ring a buzzer (CS), present food (UCS), and measure the responses (CR and UCR). Pavlov himself collected saliva with a simple measuring pouch attached to the dog's cheek; his later colleagues used a more complex device.

Pavlov found that, whenever he gave a dog food, saliva flowed in the dog's mouth. The food → salivation connection was automatic, requiring no training. Pavlov called the food the **unconditioned stimulus** and called the salivation the **unconditioned response**. The **unconditioned stimulus (UCS)** is *an event that consistently, automatically elicits an unconditioned response*, and the **unconditioned response (UCR)** is *an action that the unconditioned stimulus automatically elicits*.

Next, Pavlov introduced a new stimulus, such as a buzzer. Upon hearing the buzzer, the dog lifted its ears and looked around but did not salivate, so the buzzer was a neutral stimulus with regard to salivation. Pavlov sounded the buzzer a couple of seconds before giving food to the dog. After a few pairings of the buzzer with food, the dog began to salivate as soon as it heard the buzzer (Pavlov, 1927/1960).

We call the buzzer the **conditioned stimulus (CS)**, because the dog's *response to it depended on the preceding conditions*—that is, the pairing of the CS with the UCS. The salivation that followed the sounding of the buzzer was the **conditioned response (CR)**. The conditioned response is simply *whatever response the conditioned stimulus begins to elicit as a result of the conditioning (training) procedure*. In Pavlov's experiment and in many others, the conditioned response closely resembles the unconditioned response, but in some cases it is quite different. At the start of the conditioning procedure, the conditioned stimulus does *not* elicit a conditioned response. After conditioning, it does.

To summarize: The *unconditioned stimulus* (UCS), such as food or shock, automatically elicits the *unconditioned response* (UCR) at all times. A neutral stimulus, such as a tone or buzzer, that is paired with the UCS becomes a *conditioned stimulus* (CS). At first, this neutral stimulus elicits either no response or some irrelevant response, such as just looking around. After some number of pairings of the CS with the UCS, the conditioned stimulus elicits the *conditioned response* (CR). Figure 6.3 diagrams these relationships.

Here are some other examples of classical conditioning:

- Your alarm clock makes a faint clicking sound a couple of seconds before the alarm goes off. At first, the click by itself does not awaken you, but the alarm does. After a week or so, however, you awaken as soon as you hear the click.

Unconditioned stimulus, UCS (alarm)	→ Unconditioned response, UCR (awakening)
Conditioned stimulus, CS (clicking)	→ Conditioned response, CR (awakening)

- You hear the sound of a dentist's drill shortly before the unpleasant experience of the drill on your teeth. From then on, the sound of a dentist's drill arouses anxiety.

Unconditioned stimulus, UCS (drill on your teeth)	→ Unconditioned response, UCR (tensing the muscles)
Conditioned stimulus, CS (drill sound)	→ Conditioned response, CR (tensing the muscles)

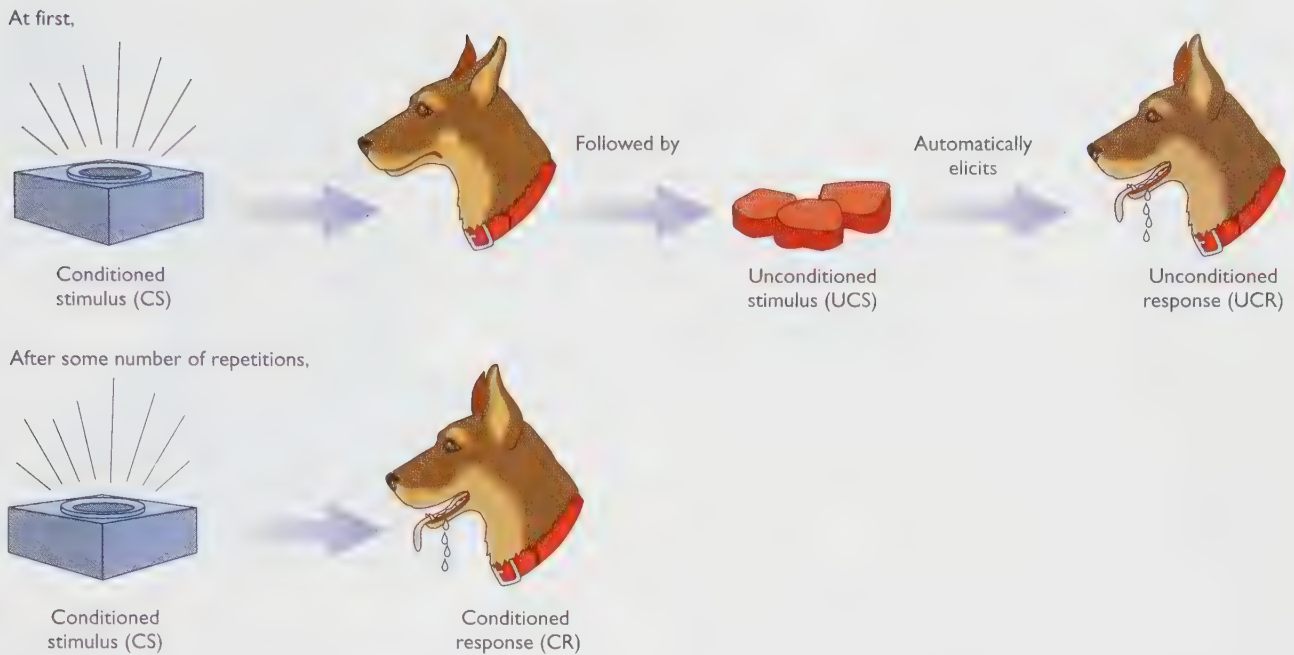


FIGURE 6.3 With classical conditioning, a conditioned stimulus is followed by an unconditioned stimulus. At first, the conditioned stimulus elicits no response, and the unconditioned stimulus elicits the unconditioned response. After sufficient pairings, the conditioned stimulus begins to elicit the conditioned response, which can resemble the unconditioned response.

- When your partner is in a romantic mood, he or she puts on some special cologne or perfume and turns on soft music. When in a grouchy mood, he or she scowls, indicating readiness for a quarrel. (Hollis, 1997, describes similar results with fish and birds. The basics of behavior apply across species.)

Unconditioned stimulus, UCS (receptive partner)	→ Unconditioned response, UCR (romantic activities)
Conditioned stimulus, CS (nice smell, soft music)	→ Conditioned response, CR (ready for romance)
Unconditioned stimulus, UCS (grouchy partner)	→ Unconditioned response, UCR (arguments)
Conditioned stimulus, CS (scowling expression)	→ Conditioned response, CR (ready for a fight)

Note the usefulness of classical conditioning in each case: It prepares an individual for likely events.

The unconditioned stimulus can be almost any stimulus that evokes an automatic response. The conditioned stimulus can be almost any detectable stimulus—a light, a sound, the cessation of a light or sound, a smell. . . . Even a mental image can be a conditioned stimulus. Psychologists in general, and behaviorists in particular, are not sure what a mental image really is, but if I tell you to “think about lemon juice,” you will salivate, and if I tell you to “think about a painful shock,” you will have some changes in breathing and heart rate. For cancer patients who have had

repeated chemotherapy treatments, thinking about chemotherapy or even imagining the building where they received it can provoke nausea and vomiting (Dadds, Bovbjerg, Redd, & Cutmore, 1997).

All else being equal, conditioning occurs more rapidly with unfamiliar than with familiar conditioned stimuli. For example, if you heard a tone a thousand times (followed by nothing) and then started hearing the tone followed by a puff of air to your left eye, you would be slow to show signs of conditioning. Similarly, imagine two people who are bitten by a snake. The one who has never been close to a snake before may develop an intense fear of snakes; the one who has spent the past 5 years tending snakes at the zoo will not develop nearly as much fear.

We shall start by discussing mostly laboratory studies but eventually come to an application of classical conditioning to the human phenomenon of drug tolerance. In Chapter 16, we shall consider the role of classical conditioning in the development of phobias.

CONCEPT CHECK

- A nursing mother consistently responds to her baby's crying by putting the baby to her breast. The baby's sucking causes the release of milk. Within a few days, as soon as the mother hears the baby crying, the milk starts to flow, even before she puts the baby to her breast. What is the conditioned stimulus? the conditioned response? the unconditioned stimulus? the unconditioned response? (Check your answers on page 204.)

The Phenomena of Classical Conditioning

The *process that establishes or strengthens a conditioned response* is known as **acquisition**. Figure 6.4 shows how the strength of a conditioned response increases after pairings of the conditioned and unconditioned stimuli. Acquisition is not the end of the story, however, because any response that can be learned can also be unlearned.

Once Pavlov had demonstrated how classical conditioning occurs, inquisitive psychologists wondered what would happen after various changes were made in the procedures. Their investigations, prompted by practical concerns, theoretical concerns, or simple curiosity, have extended our knowledge of classical conditioning. Here are a few of the main phenomena:

Extinction Suppose I sound a buzzer and then blow a puff of air into your eyes. After a few repetitions, you will start to close your eyes as soon as you hear the buzzer (Figure 6.5). Now, I sound the buzzer repeatedly without the puff of air. What do you do?

If you are like most people, you will blink your eyes the first time and perhaps the second and third times, but before long you will stop blinking. This decrease of the conditioned response is called **extinction** (see Figure 6.4). *To extinguish a classically conditioned response, repeatedly present the conditioned stimulus (CS) without the unconditioned stimulus (UCS).* That is, acquisition of a response (CR) occurs if the CS predicts the UCS; extinction occurs if the CS no longer predicts the UCS.

Be careful to distinguish between extinction and forgetting. Both serve to weaken a learned response, but they arise in different ways. Forgetting occurs when we have no opportunity to practice a certain behavior over a period of time. Extinction occurs as the result of a specific experience—the presentation of the conditioned stimulus without the unconditioned stimulus.

Extinction does not erase the original connection between the CS and the UCS, though. You might think of acquisition as learning to do something and extinction as learning to inhibit the response. For example, suppose you

have gone through original learning in which a tone regularly preceded a puff of air to your eyes. You learned to blink your eyes at the tone. Then you went through an extinction process in which you heard the tone many times but received no puffs of air. You extinguished, so that the tone no longer elicited a blink. Now, without warning, you get another puff of air to your eyes. As a result, the next time you hear the tone, you will blink your eyes! Extinction inhibited your response to the CS (here, the tone), but a sudden puff of air weakens that inhibition (Bouton, 1994).

Spontaneous Recovery Suppose that we classically condition a response and then extinguish it. Several hours or days later, we present the conditioned stimulus again. In many cases, the conditioned response will reappear. But this return is temporary, lasting only one or a few trials, unless CS-UCS pairings are resumed. **Spontaneous recovery** refers to this *temporary return of an extinguished response after a delay* (see Figure 6.4). For example, the sound of a buzzer (CS) is repeatedly followed by a puff of air blown into the eyes (UCS) until the person learns to blink at the sound of the buzzer. Then the buzzer is presented repeatedly by itself until the person learns to stop blinking. Neither the buzzer nor the puff of air is presented for the next few hours. Then the buzzer is sounded again and the person blinks—not strongly, perhaps, but more noticeably than at the end of the extinction training.

Why does spontaneous recovery take place? Think of it this way: At first, the buzzer predicted a puff of air to the eyes. Then it predicted nothing. The two sets of experiences conflict with each other, but the more recent one predominates, so the person stops blinking. Hours later, neither experience is much more recent than the other, and the effects of the original acquisition are almost as strong as those of extinction.

CONCEPT CHECK

2. In Pavlov's experiment on conditioned salivation in response to a buzzer, what procedure could you use to produce extinction? What procedure could you use to produce spontaneous recovery? (Check your answers on page 204.)

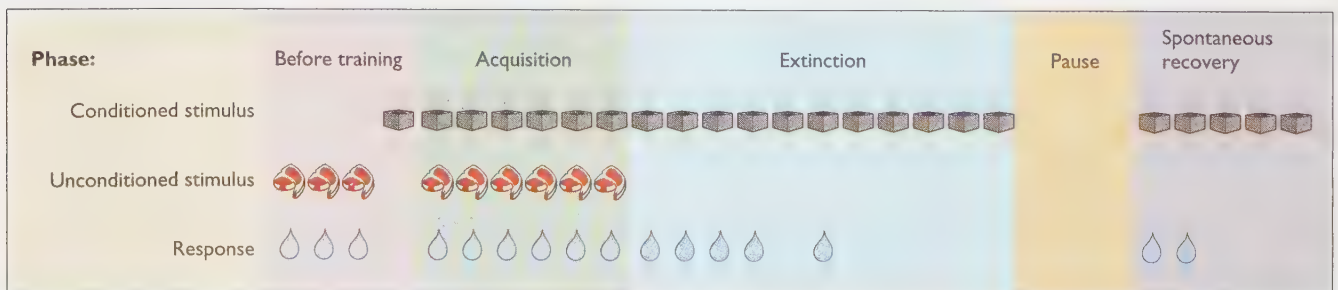


FIGURE 6.4 Phases of classical conditioning: Classical conditioning proceeds through several phases, depending on the time of presentation of the two stimuli. If the conditioned stimulus regularly precedes the unconditioned stimulus, acquisition occurs. If the conditioned stimulus is presented by itself, extinction occurs. A pause after extinction yields a brief spontaneous recovery.

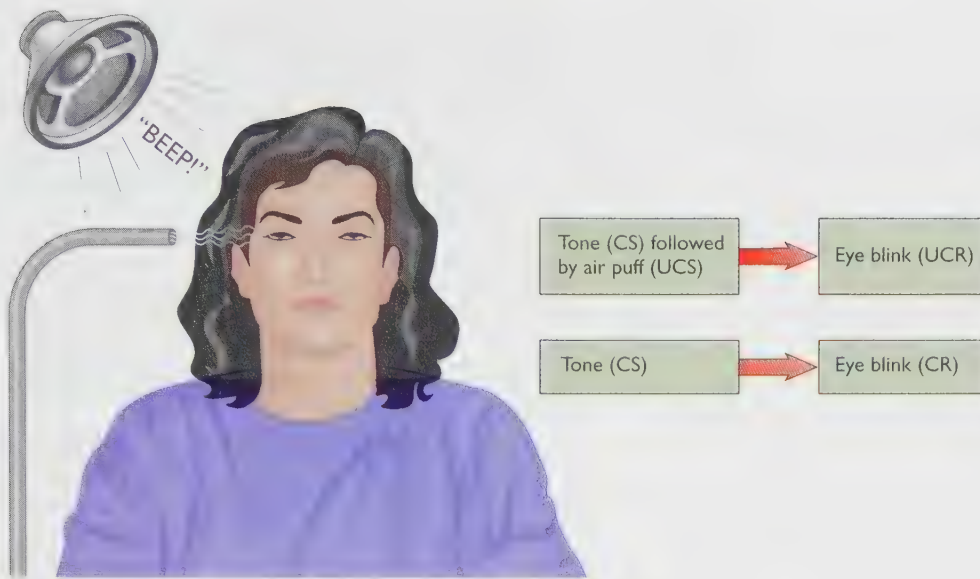


FIGURE 6.5 The procedure for classical conditioning of the eye-blink response

Stimulus Generalization

Suppose your alarm clock makes a faint clicking sound (CS) a few seconds before the alarm (UCS), and you have learned to awaken as soon as you hear the clicking sound. What happens if we now substitute a different alarm clock? It too makes a clicking sound before the alarm goes off, but it is a recognizably different click. Will it awaken you?

It probably will. The closer the sound of the new click is to the original one, the more likely you are to respond by awakening (Figure 6.6). **Stimulus generalization** is the *extension of a conditioned response from the training stimulus to similar stimuli*.

This definition may sound pretty straightforward, but in

fact psychologists find it difficult to specify exactly what “similar” means. For example, if you hear a clicking sound somewhere other than in your bedroom or at a time other than your usual time to awaken, it may not be effective at all. So, your response at any moment depends on how similar the total configuration of stimuli is to the set on which you were trained, and that similarity is hard to measure (Pearce, 1994).

Discrimination Now suppose the experimenter always follows middle C with a puff of air but never follows F-sharp with a puff of air. Eventually, you will learn to **discriminate** between these two tones: You will *respond differently to the two stimuli because different outcomes followed them*. You will blink your eyes when you hear middle C but not when you hear F-sharp. Discrimination is essential in everyday life: You learn that one bell signals that it is time for class to start, and a different bell signals a fire.

SOMETHING TO THINK ABOUT

We can easily determine how well human subjects discriminate between two stimuli. We can simply ask, “Which note has the higher pitch?” or “Which light is brighter?” How could we determine how well a nonhuman can discriminate between two stimuli? *

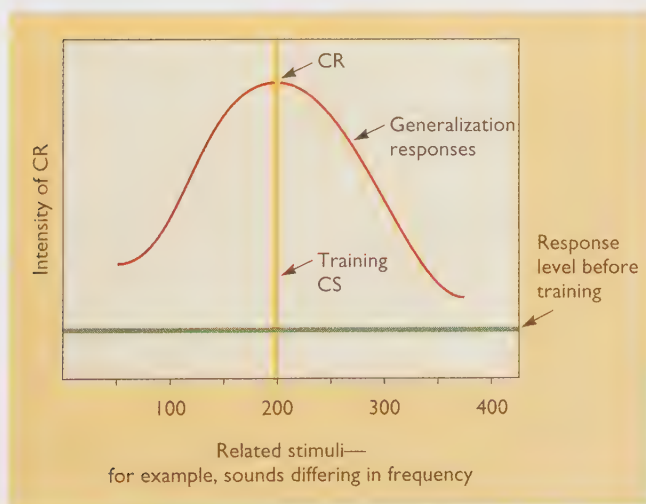


FIGURE 6.6 Stimulus generalization is the process of extending a learned response to new stimuli that resemble the one used in training. As a rule, a stimulus similar to the training stimulus elicits a strong response; a less similar stimulus elicits a weaker response.

Drug Tolerance as an Example of Classical Conditioning

Classical conditioning occurs in many laboratory settings; it also occurs in the outside world, sometimes in unexpected settings. One such setting is **drug tolerance**: *Users of certain drugs experience progressively weaker effects after taking those drugs repeatedly*. One consequence of drug tolerance is that the users crave ever larger amounts of the drug.

Drug tolerance occurs for a variety of reasons, which vary from one case to another (Poulos & Cappell, 1991). In many cases, however, drug tolerance is learned. When drug users inject themselves with morphine or heroin, the procedure is a stimulus that reliably predicts a second stimulus, the drug's entry into the brain. The drug alters experience, but it also triggers a variety of body defenses against its effects—for example, changes in hormone secretions, heart rate, and breathing rate.

Whenever one stimulus predicts a second stimulus that produces a response, the conditions necessary for classical conditioning are present. Shepard Siegel (1977, 1983) has demonstrated that classical conditioning does indeed take place during drug-injecting episodes. Initially, the injection ritual is a neutral stimulus that gives rise to no relevant response. After many pairings of that stimulus with the entry of the drug into the brain, however, the injection procedure by itself can evoke the body's antidrug defenses (Figure 6.7). From then on, the injection procedure, acting as a conditioned stimulus, triggers the defense reactions even before the drug has entered the brain and therefore weakens the effect of the drug.

If we assume that the injection procedure serves as a conditioned stimulus, then the body's defense reactions should be strongest when the drug is administered in the usual way, in the usual location, with as many familiar stimuli as possible. (The whole experience constitutes the conditioned stimulus.) The evidence strongly supports this prediction (Eikelboom & Stewart, 1982; Poulos, Wilkinson, & Cappell, 1981; Siegel, 1983; Tiffany & Baker, 1981). To show strong tolerance in a particular environment, the individual must have previously received the drug in that environment.

Why do some people die of a drug overdose that is no larger than the dose that they normally tolerated? According to

the classical-conditioning interpretation, these people probably took the fatal overdose in an unfamiliar setting. For example, someone who is accustomed to taking a drug at home in the evening could suffer a fatal reaction from taking it at a friend's house in the morning. Because the new setting did not serve as a CS, it failed to trigger the usual drug tolerance.

CONCEPT CHECKS

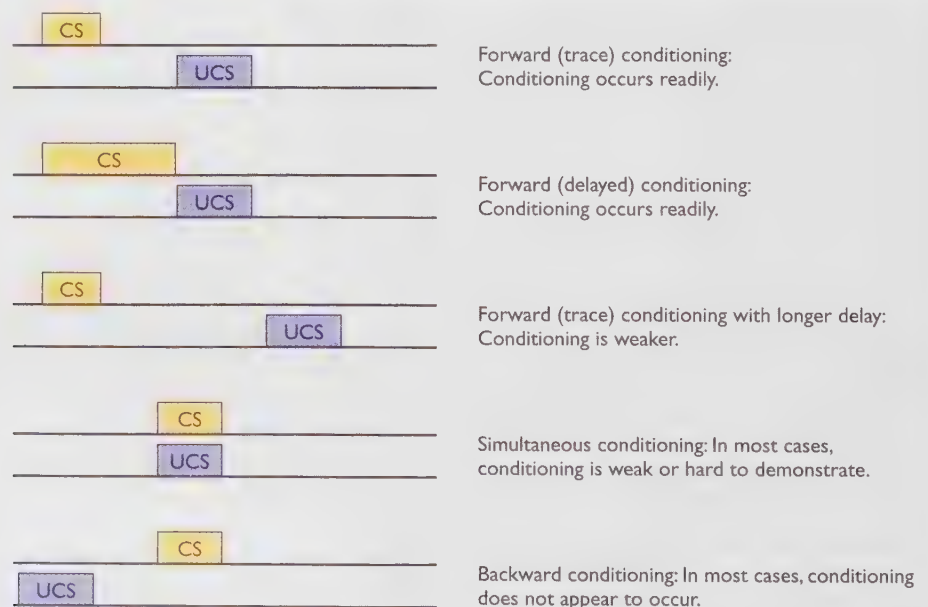
3. When an individual develops tolerance to the effects of a drug injection, what are the conditioned stimulus, the unconditioned stimulus, the conditioned response, and the unconditioned response?
4. Within the classical-conditioning interpretation of drug tolerance, what procedure should extinguish tolerance? (Check your answers on page 204.)

Explanations of Classical Conditioning

What is classical conditioning, really? As is often the case, the process appeared at first to be fairly simple, but later investigation found it to be a more complex, and perhaps more interesting, phenomenon.

Pavlov's Theory of the Causes of Classical Conditioning

Pavlov believed that, for classical conditioning to occur, the conditioned stimulus and the unconditioned stimulus must be close together in time, with the conditioned stimulus occurring first. In the following sketches, read time left to right:



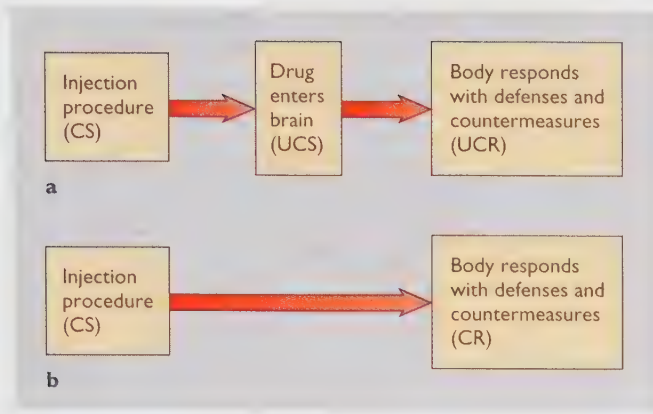


FIGURE 6.7 If a particular injection procedure consistently predicts the entry of a drug into the brain, an individual can develop a conditioned response in defense against the drug. This conditioned response is an important part of drug tolerance.

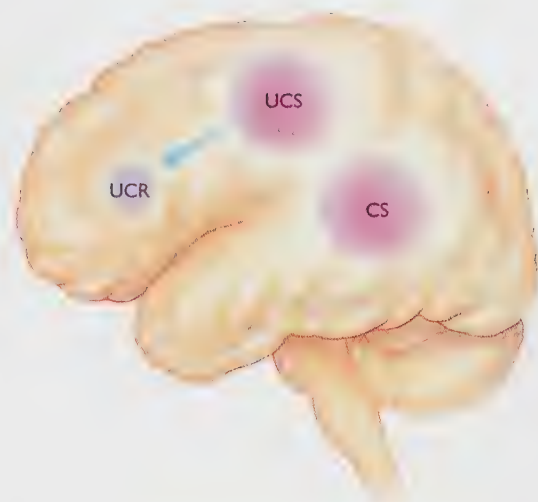
Conditioning occurs most readily with forward conditioning, with the CS and UCS close in time. *Nearness in time* is called **temporal contiguity**. With rare exceptions, the conditioned stimulus must be presented first, followed quickly by the unconditioned stimulus. In some cases, the conditioned stimulus (such as a buzzer) continues until the presentation of the unconditioned stimulus; in other cases, the conditioned stimulus stops before the unconditioned stimulus does. In either case, however, the delay is short between the start of one stimulus and the start of the other. All else being equal, the longer the delay between the CS and the UCS, the weaker the conditioning will be.

Pavlov proposed a neurological theory, assuming that every stimulus excites a specific area of the brain. A buzzer excites a “buzzer center,” and meat excites a “meat center.” Exciting both centers at the same time establishes and strengthens a connection between them. From then on, any excitation of the buzzer center (CS) also excites the meat center (UCS) and evokes salivation (Figure 6.8).

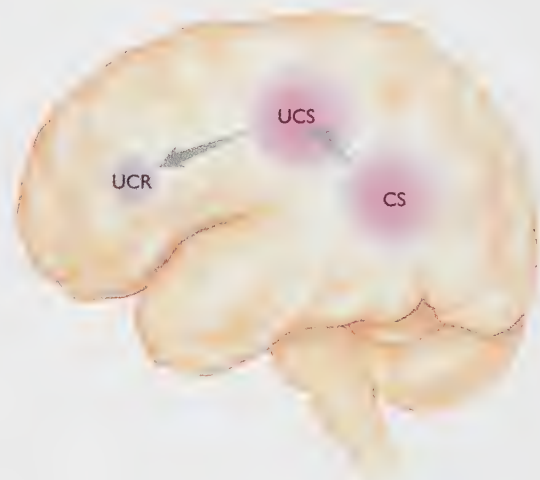
Pavlov’s theory was appealing because it offered a simple, mechanical explanation of learning, even though he had no evidence for the existence of CS centers or UCS centers, much less connections between them. Later evidence has demonstrated the need for a more complex explanation, however.

A Signal, Not Just a Transfer of Responses

According to Pavlov’s view of classical conditioning, an animal comes to respond to the conditioned stimulus as if it were the unconditioned stimulus. With his results, that interpretation was reasonable; the conditioned response and the unconditioned response were both salivation. However, in some situations the conditioned and unconditioned responses can be quite different. For example, a shock (UCS) causes rats to jump and shriek, but a conditioned stimulus predicting the shock makes rats freeze in position. They do



a



b

FIGURE 6.8 Pavlov believed that conditioning depended on temporal contiguity: (a) At the start of conditioning, activity in the UCS center automatically causes activation of the UCR center. At this time, activity of the CS center does not affect the UCS center. (b) After sufficient pairings of the CS and UCS, their simultaneous activity causes the growth of a connection between the CS and UCS centers. Afterward, activity in the CS center will flow to the UCS center and therefore excite the UCR center.

not react to the conditioned stimulus as if it were a shock but as they do to signals of possible danger. In short, the conditioned response serves to prepare the individual for the unconditioned stimulus.

WHAT IS THE EVIDENCE?

Contiguity Alone or Contingency

Contrary to what Pavlov believed, temporal contiguity is not always sufficient to establish classical conditioning. We shall consider two highly influential experiments.

EXPERIMENT 1

Hypothesis For this study, we compare two hypotheses: One is that pairing a new stimulus repeatedly with a shock will produce a conditioned response to that stimulus. The other hypothesis is that conditioning to this stimulus will fail if another stimulus already predicted the shock.

Method With one group of rats, a light (CS) was repeatedly followed by a shock (UCS) until the rats showed a clear, consistent response to the light. With a second group, a tone (CS) was followed by the shock until the rats consistently responded to the tone. Then, both groups experienced the light and the tone simultaneously, followed by the same shock. Later, the experimenter tested the rats' reactions to the light and the tone, presented separately (Kamin, 1969). (See Figure 6.9.)

Results After pairing of the combined light-plus-tone with shock, rats continued to respond as before to whichever stimulus they had originally associated with shock (light for one group, tone for the other). However, they responded very weakly to the new, added stimulus. That is, even though the new stimulus was always followed by the shock, animals developed little response to it. These results demonstrate the **blocking effect**: *The previously established association to one stimulus blocks the formation of an association to the added stimulus.*

Interpretation If temporal contiguity were the only factor responsible for learning, the rats should have learned a strong response to both the light and the tone, because both were presented just before the shock. The failure of the rats to learn a response to the new stimulus indicates that con-

ditioning depends on more than just presenting two stimuli together in time; the first stimulus must be informative or predictive of the second stimulus.

EXPERIMENT 2

Hypothesis Again, we compare two hypotheses. The first is that conditioning will occur whenever a CS is consistently followed by a UCS. The second is that conditioning will occur only if the UCS is likely after the CS *and unlikely without the CS*.

Method For rats in Group 1, conditioned stimulus and unconditioned stimulus were presented in the sequence shown at the top of Figure 6.10. The horizontal line represents time; the vertical arrows represent times of stimuli presentation. For rats in Group 2, the two stimuli were presented in the sequence shown at the bottom of Figure 6.10. In both cases, every presentation of the conditioned stimulus immediately preceded a presentation of the unconditioned stimulus. But in the second case, the unconditioned stimulus occurred frequently both in the presence and in the absence of the conditioned stimulus; therefore, the CS was a poor predictor of the UCS (Rescorla, 1968, 1988).

Results Rats receiving the first sequence of stimuli developed a strong response to the conditioned stimulus. Those receiving the second sequence of stimuli developed little or no response (Rescorla, 1968, 1988).

Interpretation Although both groups of rats received the same number of CS-UCS pairings, one group learned a response to the CS and the other group did not. Evidently, *animals (including humans) associate a conditioned stimulus*

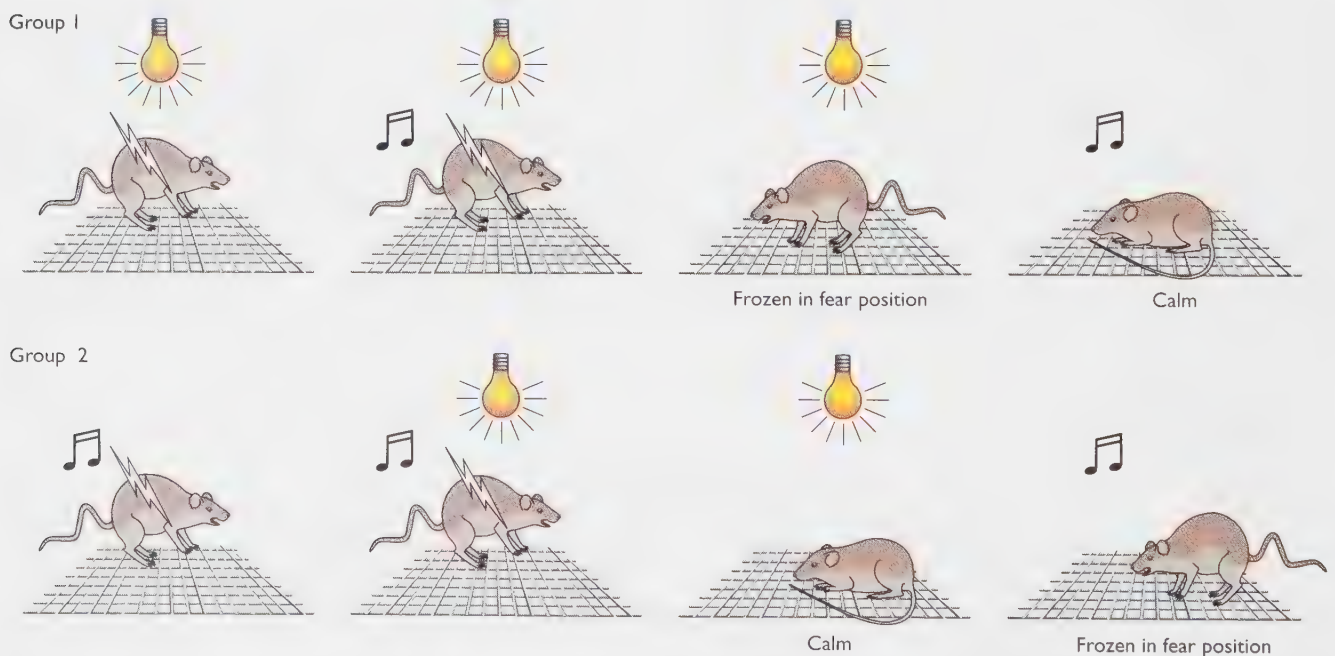


FIGURE 6.9 In Kamin's experiment, each rat first learned to associate either light or sound with shock. Then, it received a compound of both light and sound followed by shock. Even after many pairings, each rat continued to show fear of its old stimulus (the one that already predicted shock). The rats showed little response to the new stimulus.

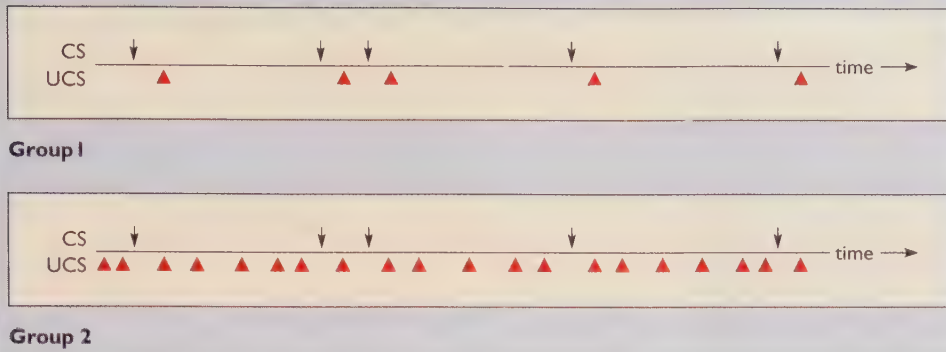


FIGURE 6.10 In Rescorla's experiment, the CS was always followed by the UCS in both groups. However, Group 2 received the UCS so frequently at other times that the CS was not a useful predictor. Group 1 developed a strong conditioned response to the CS; Group 2 did not.

with an unconditioned stimulus only when the CS predicts the occurrence of the UCS. If the conditioned stimulus immediately precedes the unconditioned stimulus but provides no new information, it is ineffective for conditioning.

* * *

CONCEPT CHECKS

5. If temporal contiguity were the only factor responsible for classical conditioning, what result should the experimenters have obtained in Experiment 2?
6. Suppose you have already learned to flinch when you hear the sound of a dentist's drill, because of the association between that sound and forthcoming pain. Now your dentist turns on some soothing background music at the same time as the drill. That is, the background music is paired with the pain just as much as the drill sound is. Will you learn to flinch at the sound of that background music, if it is presented by itself? (Check your answers on page 204.)

Conditioning, Contiguity, and Contingency

A conditioned response develops only if there is a **contingency** (*predictability*)—that is, only if the UCS is more likely after the CS than it would be otherwise. Classical conditioning has therefore been compared to scientific reasoning (Denniston, Miller, & Matute, 1996): The learner discovers which event predicts which outcome.

However, we should not imagine that a rat literally calculates the probability of a shock after a tone versus the probability of a shock during a period without a tone (Papini & Bitterman, 1990). Indeed, there is no reason to assume that the animal is “thinking” about these events at all. Even in humans, it is possible to produce conditioning when people are not consciously aware of the stimuli. In one study, people looked at photos of faces presented “subliminally” (briefly, followed by an interfering stimulus). One photo was always followed by shock. Even after people developed a conditioned stimulus to that photo, though, they could not describe it. And when they finally got a good, long look at the photo, they described it as unfamiliar (Parra, Esteves, Flykt, & Öhman, 1997).

Therefore, when we say that the results of classical conditioning are *like* scientific reasoning, we mean that both produce an outcome that depends on contingencies, or predictabilities. The underlying mechanisms do not require deep thinking; they might only be combinations of relatively simple processes.

THE MESSAGE

Classical Conditioning Is More than Drooling Dogs

People sometimes use the term “Pavlovian” to mean simple, mechanical, robot-like behavior. But Pavlovian or classical conditioning is not a mark of stupidity: It is a way of responding to relationships among events, a way of preparing us for what is likely to happen. Classical conditioning requires the processing of a fair amount of information.

Classical conditioning is important for some aspects of our behavior but less important for others. It alters our motivational or emotional reactions to stimuli, our “gut feelings”—including responses related to fear, preparations for eating, preparations for a drug injection, and so forth. But it does not control walking toward or away from various stimuli. That is, classical conditioning might tell us to be afraid, but it does not tell us how to avoid the frightening item. It might tell us to salivate in preparation for eating, but it does not tell us how to find food. Other types of learning can answer these other questions, as we shall see later in this chapter.

SUMMARY

* **Classical conditioning.** Ivan Pavlov discovered classical conditioning, the process by which an organism learns a new association between two stimuli that have been paired with each other—a neutral stimulus (the conditioned stimulus) and one that initially evokes a reflexive response (the unconditioned stimulus). The organism displays this association by responding in a new way (the conditioned response) to the conditioned stimulus. (page 195)

* **Extinction.** After classical conditioning has established a conditioned response to a stimulus, the response

can be extinguished by repeatedly presenting that stimulus by itself. (page 198)

★ **Spontaneous recovery.** If the conditioned stimulus is not presented at all for some time after extinction and is then presented again, the conditioned response may return to some degree. That return is called spontaneous recovery. (page 198)

★ **Stimulus generalization.** An individual who learns to respond to one stimulus will respond similarly to similar stimuli. However, it is difficult to specify how we should measure similarity. (page 199)

★ **Discrimination.** If one stimulus is followed by an unconditioned stimulus and another similar stimulus is not, the individual will come to discriminate between these two stimuli. (page 199)

★ **Drug tolerance.** Drug tolerance is partly a form of classical conditioning in which the drug administration procedure becomes associated with the effects of the drug. (page 199)

★ **Temporal contiguity versus contingency.** Pavlov believed that temporal contiguity between two stimuli caused classical conditioning. Later studies indicate that conditioning depends also on contingency, or the extent to which the occurrence of the first stimulus predicts the occurrence of the second. (page 200)

Suggestion for Further Reading

Rescorla, R. A. (1988). Pavlovian conditioning: It's not what you think it is. *American Psychologist*, 43, 151–160. A theoretical review by an investigator who has contributed significantly to changing views of classical conditioning.

Terms

unconditioned reflex an automatic connection between a stimulus and a response (page 196)

classical conditioning or **Pavlovian conditioning** the process by which an organism learns a new association between two paired stimuli—a neutral stimulus and one that already evokes a reflexive response (page 196)

unconditioned stimulus (UCS) a stimulus that automatically elicits an unconditioned response (page 196)

unconditioned response (UCR) an automatic response to an unconditioned stimulus (page 196)

conditioned stimulus (CS) a stimulus that comes to evoke a particular response after being paired with the unconditioned stimulus (page 196)

conditioned response (CR) Whatever response the conditioned stimulus begins to elicit as a result of the conditioning procedure (page 196)

acquisition the process by which a conditioned response is established or strengthened (page 198)

extinction in classical conditioning, the dying out of the conditioned response after repeated presentations of the conditioned stimulus without the unconditioned stimulus (page 198)

spontaneous recovery the temporary return of an extinguished response after a delay (page 198)

stimulus generalization the extension of a conditioned response from the training stimulus to similar stimuli (page 199)

discrimination making different responses to different stimuli that have been followed by different outcomes (page 199)

drug tolerance the progressively weaker effects of a drug after repeated use (page 199)

temporal contiguity nearness in time (page 201)

blocking effect the tendency of a previously established association to one stimulus to block the formation of an association to an added stimulus (page 202)

contingency the prediction of one stimulus from the presence of another (page 203)

Answers to Concept Checks

1. The conditioned stimulus is the baby's crying. The unconditioned stimulus is the baby's sucking at the breast. Both the conditioned response and the unconditioned response are the release of milk. Many nursing mothers experience this classically conditioned reflex. (page 197)
2. To bring about extinction, present the buzzer repeatedly without presenting any food. To bring about spontaneous recovery, first bring about extinction; then wait hours or days and present the buzzer again. (page 198)
3. The conditioned stimulus is the injection procedure. The unconditioned stimulus is the entry of the drug into the brain. Both the conditioned response and the unconditioned response are the body's defenses against the drug. (page 200)
4. To extinguish tolerance, present the injection procedure (conditioned stimulus) without injecting the drug (unconditioned stimulus). Instead, inject just water or salt water. Shepard Siegel (1977) demonstrated that repeated injections of salt water do reduce tolerance to morphine in rats. (page 200)
5. If temporal contiguity were the only factor responsible for classical conditioning, the rats exposed to the first sequence of stimuli should have responded just as those exposed to the second sequence of stimuli. In both cases, the CS was always followed by the UCS. (page 203)
6. No, you will not learn to flinch at the sound of the background music. Because the drill sound already predicted the pain, the new stimulus is uninformative and will not be strongly associated with the pain. Your results will be an example of the blocking effect. (page 203)

Web Resources

Factors Determining the Effectiveness of Classical Conditioning

www.biozentrum.uni-wuerzburg.de/~brembs/classical/classical.html

This fairly technical discussion also covers the intricacies of learned associations.

Operant Conditioning

How do the consequences of our behaviors affect future behaviors?

Sometimes a very simple idea can be amazingly powerful. Consider democracy, for example: What could be simpler than the idea that every person gets one vote? Or consider the idea of natural selection, an extremely simple concept that brings order to an enormous array of biological facts that would otherwise seem unrelated. In this module we shall consider the idea that behaviors become more likely or less likely because of their consequences. In other words, we either repeat a behavior or cease doing it, depending on the outcome. This simple, even obvious, idea is quite powerful.

Thorndike and Operant Conditioning

Shortly before Pavlov performed his innovative experiments, Edward L. Thorndike (1911/1970), a Harvard graduate student, had begun to train and test some cats in a basement. Saying that earlier experiments had dealt only with animal intelligence, never with animal stupidity, he devised a simple, behavioristic explanation of learning.

Thorndike put cats into puzzle boxes (Figure 6.11) from which they could escape by either pressing a lever, pulling a string, or tilting a pole. Sometimes he placed food outside the box. (Usually, though, just escaping from the small box was reinforcement enough.) The cats learned to make whatever response opened the box. Thorndike discovered that they learned faster if the response opened the box immediately; any delay would impair learning.

When a cat had to tilt a pole to escape from the box, it would first paw or gnaw at the door, scratch the walls, and pace back and forth. Eventually, it would bump against the pole by accident and the door would open. The next time, the cat would go through the same repertoire of behaviors but might bump against the pole a little sooner. Over many trials, the time it took the cat to escape grew shorter, in a

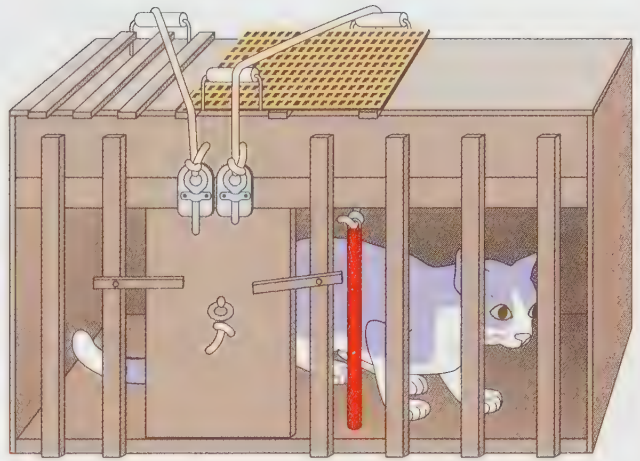


FIGURE 6.11 Each of Thorndike's puzzle boxes had a device that could open it. Here, tilting the pole will open the door. (Based on Thorndike, 1911/1970.)

gradual and irregular fashion. Figure 6.12 shows a learning curve to represent this behavior. A *learning curve* is a graph of the changes in behavior that occur over successive trials in a learning experiment.

Had the cat “figured out” how to escape? Had it come to “understand” the connection between bumping against the pole and opening the door? No, said Thorndike. If the cat had gained some new insight at some point along the way, he explained, its speed of escaping would have suddenly increased at that point. Actually, the cat's performance improved slowly, gradually, and inconsistently. One could not designate at which point the cat understood.

Thorndike concluded that learning occurs only when certain behaviors are strengthened at the expense of others. An animal enters a given situation with a certain repertoire of responses—pawing the door, scratching the walls, pacing,

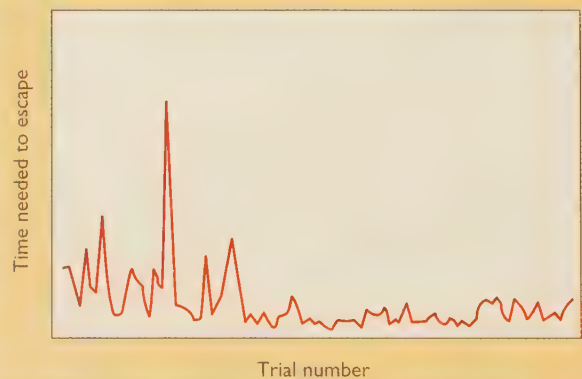


FIGURE 6.12 Trial and error or insight? As the data from one of Thorndike's experiments show, the time that a cat needs to escape from a puzzle box gradually grows shorter, but in an irregular manner. Thorndike concluded that the cat did not at any point “suddenly get the idea.” Instead, reinforcement gradually increased the probability of the successful behavior.

and so forth (labeled R_1 , R_2 , R_3 , . . . in Figure 6.13). First, the animal engages in its most probable response for this situation (response R_1 in the figure). If nothing special happens, it proceeds to other responses. Eventually, it attempts a lower-probability response—for example, bumping against the pole—that opens the door (response R_7 in the figure). The opening of the door serves as a reinforcement.

A **reinforcement** is *an event that increases the future probability of the most recent response*. Thorndike said that it “stamps in,” or strengthens, the response. The next time Thorndike’s cat is in the puzzle box, it may have a slightly higher probability of bumping the lever; after another reinforcement, the probability will go up another notch. Eventually, the pole-bumping response becomes the most probable response in this situation, and the cat escapes quickly (Figure 6.13c).

Thorndike summarized his views in the **law of effect** (Thorndike, 1911/1970, p. 244): “Of several responses made to the same situation, those which are accompanied or closely followed by satisfaction to the animal will, other things being equal, be more firmly connected with the situation, so that, when it recurs, they will be more likely to recur.” In other words, the animal becomes more likely to repeat the responses that led to favorable consequences. This process does not require that the animal “think” or “understand,” however. A fairly simple machine could produce responses at random and then repeat the ones that led to reinforcement.

The process of changing behavior by following a response with reinforcement is known as **operant conditioning** (because the subject operates on the environment to produce an outcome) or **instrumental conditioning** (because the subject’s behavior is *instrumental* in producing the outcome). The defining difference between operant conditioning and classical conditioning is one of procedure: *In operant conditioning, the subject’s behavior determines an outcome and is affected by that outcome. In classical conditioning, the subject’s behavior has no effect on the outcome (the presentation of either the CS or the UCS).*

In general, the two kinds of conditioning also differ in the behaviors that they affect. That is, classical conditioning applies primarily to **visceral responses** (*responses of the internal organs*), such as salivation and digestion, whereas operant conditioning applies primarily to **skeletal responses**—that is, *movements of leg muscles, arm muscles, and so forth*. However, this distinction sometimes breaks down. For example, if a tone is consistently followed by an electric shock (a classical-conditioning procedure), the tone will make the animal freeze in position (a skeletal response) as well as increase its heart rate (a visceral response).

CONCEPT CHECK

- When I ring a bell, an animal sits up on its hind legs and drools; then I give it some food. Is the animal’s behavior an example of classical conditioning or of operant conditioning?

So far, you do not have enough information to answer the question. What else would you need to know before you could answer? (Check your answer on page 218.)

Extinction, Generalization, Discrimination, and Discriminative Stimuli

No doubt you are familiar with the saying, “If at first you don’t succeed, try, try again.” Better advice is to try again, but differently. If a particular action fails or used to succeed but is now failing, perhaps you should quit that action and try something else instead.

In operant conditioning, **extinction occurs if responses stop producing reinforcements**. For example, you were once in the habit of asking your roommate to join you for supper. The last five times you asked, your roommate said no, so you stop asking. In classical conditioning, extinction is achieved by presenting the CS without the UCS. Table 6.1 compares operant conditioning and classical conditioning.

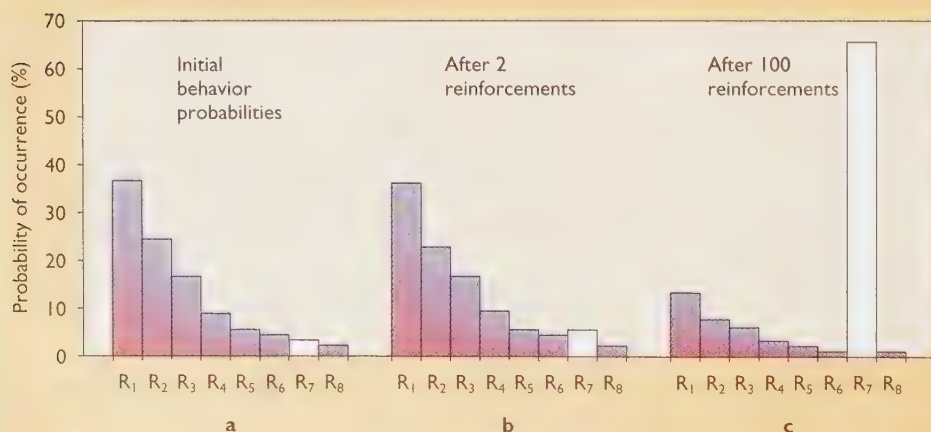


FIGURE 6.13 According to Thorndike, a cat starts with a large set of potential behaviors in a given situation. When one of these, such as pushing at a door, leads to reinforcement, the future probability of that behavior increases. We do not need to assume that the cat understands what it is doing or why.

TABLE 6.1 Comparison of Classical Conditioning and Operant Conditioning

	CLASSICAL CONDITIONING	OPERANT CONDITIONING
Terminology	CS, UCS, CR, UCR	Response, reinforcement
Subject's behavior ...	Does not control UCS	Controls reinforcement
Paired during acquisition	Two stimuli (CS and UCS)	Response and reinforcement (in the presence of certain stimuli)
Responses studied	Mostly visceral (internal organs)	Mostly skeletal (movements)
Extinction procedure	CS without UCS	Response without reinforcement

You will recall the phenomena of stimulus generalization and discrimination in classical conditioning. Similar phenomena occur in operant conditioning. An individual who receives reinforcement for a particular response to a certain stimulus will probably make the same response to a similar stimulus. *The more similar a new stimulus is to the original reinforced stimulus, the more strongly the subject is likely to respond.* This phenomenon is known as **stimulus generalization**. For example, you might reach for the turn signal in a rented car in the same place where you would find it in your own car.

An individual who is *reinforced for responding to one stimulus and not for responding to another stimulus* will come to **discriminate** between them and *will respond more vigorously to one than to the other*. For example, you walk toward a parked car that you think is yours but then you realize it is not. After several such experiences, you learn to identify your own car from a distance.

A *stimulus that indicates which response is appropriate or inappropriate* is called a **discriminative stimulus**. A great deal of our behavior is governed by discriminative stimuli. For example, your professor standing at the front of the class is a discriminative stimulus to stop talking and get ready to take notes. When class is about over, the hands on the clock provide a discriminative stimulus to get ready to leave the room. A scowl on your roommate's face is a discriminative stimulus for you to keep quiet. Road signs provide discriminative stimuli to tell you when to speed up, slow down, or change lanes. Throughout your day, one stimulus after another signals which behaviors are likely to be reinforced and which ones are not.

Why Are Certain Responses Learned More Easily Than Others?

Thorndike's cats quickly learned to push and pull various devices in their efforts to escape from his puzzle boxes. But when Thorndike tried to teach them to scratch or lick themselves to receive the same reinforcement, they learned slowly and performed inconsistently. Why?

One possible reason is **belongingness**, the *concept that certain stimuli "belong" together or that a given response is more readily associated with certain outcomes*

than with others. Belongingness is an idea that Thorndike himself suggested, although psychologists neglected it for decades, preferring to believe that animals could just as easily associate almost any stimulus with any response. Eventually, psychologists revived the concept of belongingness, also sometimes known as "preparedness" (Seligman, 1970). For example, dogs can readily learn that a sound coming from one location means "raise your left leg," and a sound coming from another location means "raise your right leg." But it takes them virtually forever to learn that a ticking metronome means raise the left leg and a buzzer means raise the right leg (Dobrzecka, Szwejkowska, & Konorski, 1966). (See Figure 6.14.)

Presumably, Thorndike's cats were slow to associate scratching themselves with escaping from a box because the two activities do not "belong" together. (Cats evolved the ability to learn "what leads to what" in the real world, and scratching oneself is very unlikely to open doors in the real world.) But there is another possible explanation for why cats have trouble learning to scratch themselves for reinforcement: Perhaps a cat will scratch itself only when it itches (Charlton, 1983). Consider what would happen if you knew that you could win a large prize if you finished first in a rapid swallowing contest. (Why not? People compete at everything else.) You quickly swallow once, twice, maybe three times, but each successive swallow gets harder and harder. (Go ahead and try it.) Some behaviors are just not easy to produce in large quantities.

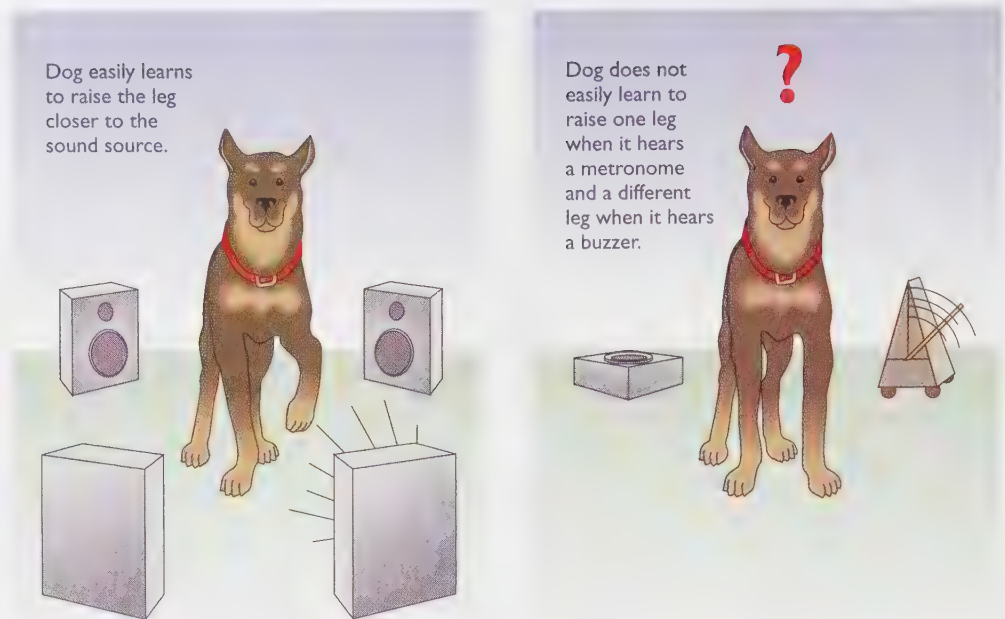
TRY IT YOURSELF

B. F. Skinner and the Shaping of Responses

The most influential radical behaviorist, B. F. Skinner (1904–1990), demonstrated many uses of operant conditioning. Skinner was an ardent practitioner of parsimony, always seeking simple explanations in terms of reinforcement histories rather than more complex explanations in terms of mental states.

One problem confronting any student of behavior is how to define a response. For example, imagine watching a group of children and trying to count "aggressive behaviors."

FIGURE 6.14 According to Thorndike's principle of belongingness, some items are easy to associate with each other because they "belong" together; others do not. For example, dogs easily learn to use the direction of a sound as a signal for which leg to raise, but they have trouble using the *type* of sound as a signal for which leg to raise.



What is an aggressive act and what isn't? Psychologists studying intelligence, emotion, or personality spend much of their time trying to find the best method of measurement. Even simple food-getting or shock-escaping responses are hard to define and measure. Skinner dispensed with the arguments (Zuriff, 1995): He set up a box, called an *operant chamber* (or *Skinner box*, a term that Skinner himself never used), in which a rat presses a lever or a pigeon pecks an illuminated disk, or "key," to receive food (Figure 6.15). He then operationally defined the response as anything that the animal did to depress the lever or key. So, if the rat pressed the lever with its snout instead of its paw, the response still counted; if the pigeon batted the key with its wing instead of pecking it with its beak, it still counted. The behavior was defined by its outcome, not the muscle movements.

Does that definition make sense? Skinner's reply was that it did, because it led to consistent results in his experiments. Many other researchers agreed, and Skinner's procedures became standard in many laboratories. When deciding how to define a term (such as *response*), the best definition is the most useful one. If our purposes change, we can change the definition later.

Shaping Behavior

Suppose that you want to train a rat to press a lever. If you simply put the rat in a box and wait, the rat might never press the lever. To avoid interminable waits, Skinner introduced a powerful technique, called **shaping**, for *establishing a new response by reinforcing successive approximations to it*.

To *shape* a rat to press a lever, you might begin by reinforcing the rat for standing up, a common behavior in rats. Before long, the rat has received several reinforcements and is beginning to stand up more frequently. Now you change

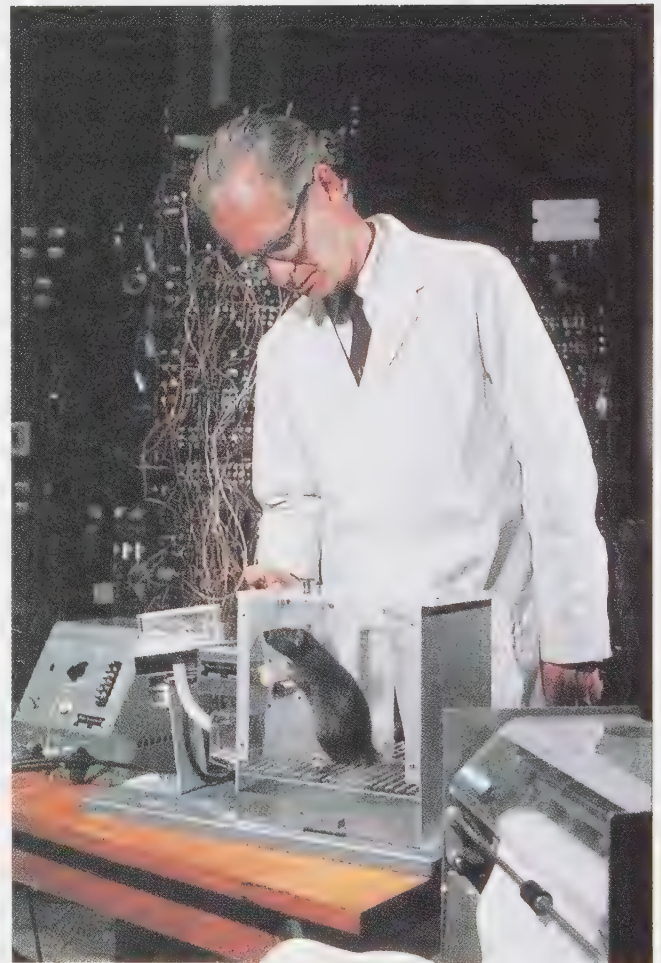


FIGURE 6.15 B. F. Skinner examines one of his laboratory animals in an operant-conditioning chamber, or "Skinner box." When the light above the bar is on, pressing the bar is reinforced. A food pellet rolls out of the storage device (left) and down the tube into the cage.

the rules, giving food only when the rat stands up while facing in the general direction of the lever. Soon, the rat spends much of its time standing up and facing the lever. (It extinguishes its behavior of standing and facing in any other direction, because those responses are not reinforced.) Next, you provide reinforcement only when the rat stands facing in the right direction and in the half of the cage nearest the lever. You gradually move the boundary, and the rat moves closer to the lever. Then the rat must touch the lever and, finally, apply weight to it. Through a series of short, easy steps, you might thus shape the rat to press levers in a matter of minutes. Similarly, education is a kind of shaping procedure: First, your parents or teachers praise you for counting your fingers; later, you must add and subtract to earn their congratulations; step by step, your tasks get more complex, until you are doing calculus.

Chaining Behavior

To produce complex sequences of behavior, psychologists use a procedure called **chaining**. Assume that you want to train an animal, perhaps a guide dog or a show horse, to go through a sequence of actions in a particular order. You could *chain* the behaviors, *reinforcing each one with the opportunity to engage in the next behavior*. That is, first

the animal learns the final behavior for a reinforcement; then it learns the next-to-last behavior, which is reinforced by the opportunity to perform the final behavior. And so on.

For example, a rat might first be placed on the top platform in Figure 6.16, where it eats food. Then it is placed on the intermediate platform with a ladder in place leading to the top platform. The rat learns to climb the ladder. After it has done so repeatedly, it is placed again on the intermediate platform but this time the ladder is not present. It must learn to pull a string to raise the ladder so that it can climb to the top platform. Finally, the rat is placed on the bottom platform. It now has to learn to climb the ladder to the intermediate platform, pull a string to raise the ladder, and then climb the ladder again. For each response in the chain, the reinforcement is the opportunity to engage in the next behavior; the final response in the chain leads to food.

Humans learn to make chains of responses, too. As an infant you learned to eat with a fork and spoon. Later, you learned to put your own food on the plate before eating. Eventually, you learned to plan a menu, go to the store, buy the ingredients, cook the meal, put it on the plate, and then eat it. Each behavior is reinforced by the opportunity to engage in the next behavior.

To show how effective shaping and chaining can be, Skinner sometimes performed this demonstration: First, he

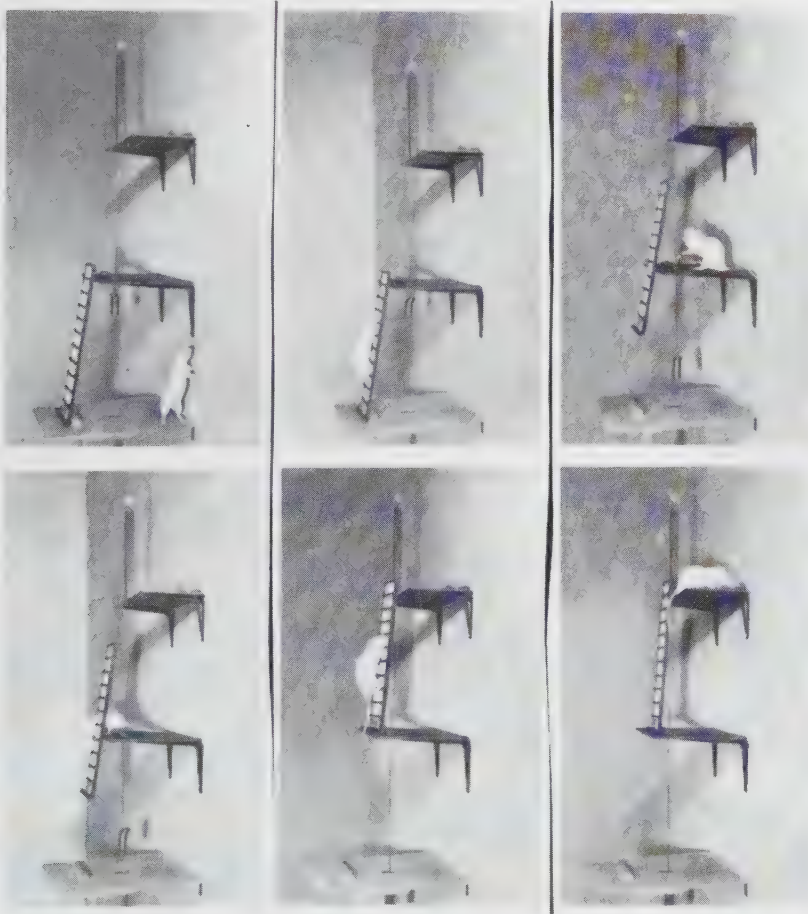


FIGURE 6.16 Chaining is a procedure in which the reinforcement for one behavior is the opportunity to engage in the next behavior. To reach food on the top platform, this rat must climb a ladder and pull a string to raise the ladder so that it can climb up again. Behavior chains longer than this can be sustained by reinforcement at the end of the chain.

TABLE 6.2 Four Categories of Operant Conditioning

	ITEM SUCH AS FOOD OR PRAISE	ITEM SUCH AS PAIN OR DISCOMFORT
Behavior produces the item.	Positive reinforcement (behavior increases in frequency) Example: You write a short story and receive a prize for it. (You learn to write more short stories and submit them in contests.) 	Punishment , also called <i>passive avoidance</i> (behavior decreases in frequency) Example: You insult someone who then <i>slaps</i> your face. (You learn not to insult people.) 
Behavior prevents or removes the item.	Omission training , also called <i>punishment or negative punishment</i> (behavior decreases in frequency) Example: You eat too much in winter and then in spring you can't fit into your swimsuit. (You learn not to overeat.) 	Escape or active avoidance , also called <i>negative reinforcement</i> (behavior increases in frequency) Example: You carry an umbrella with you all day and as a result you <i>avoid</i> getting wet when the rains come. (You learn to carry your umbrella when you hear a forecast of rain.) 

trained a rat to go to the center of a cage. Then he trained it to do so only when he was playing a certain record. Then he trained it to wait for the record to start, go to the center of the cage, and sit up on its hind legs. Step by step, Skinner eventually trained the rat to wait for the record to start (which happened to be the "Star-Spangled Banner"), move to the center of the cage, sit up on its hind legs, put its claws on a string next to a pole, pull the string to hoist a flag, and then salute the flag until the record had finished. Only then did the rat get its reinforcement. Needless to say, a show of patriotism is not part of a rat's natural repertoire of behavior; it learns to go through the motions only by successive approximations.

Increasing and Decreasing the Frequency of Responses

Nearly all of our behavior is governed by its consequences. Investigators of operant conditioning try to determine in detail how those consequences exert their effects.

Reinforcement and Punishment

Recall that *reinforcement* is an event that increases the probability that a response will be repeated. A **punishment** is an event that decreases the probability of a response. A reinforcement can be either the presentation of an item such as food or the removal or avoidance of an item such as pain. A punishment can be either the presentation of an item such as pain or the removal of an item such as

food. One is tempted to say that reinforcement is a pleasant event and punishment an unpleasant event, but we do not really know what rats, pigeons, and other species experience. Moreover, addicts work extremely hard for alcohol, drugs, and gambling activity, even when they seem to receive virtually no pleasure from these things (Berridge & Robinson, 1995). So, we should not equate reinforcement with pleasure.

Table 6.2 outlines the four possibilities for reinforcement and punishment and introduces some terms, including a couple that are potentially confusing. Let's go through these terms and procedures carefully. **Positive reinforcement** (on the upper left of the table) is the *presentation of an event that strengthens or increases the likelihood of a behavior*. Examples are food, water, or access to a sexual partner.

Punishment (upper right) occurs when a response is followed by an event such as pain. For example, you put your hand on a hot stove and burn yourself; you insult someone and that person slaps you. Punishment is also called **passive avoidance learning** because *the individual learns to avoid a particular outcome by being passive* (for example, by *not* putting your hand on the stove or by *not* insulting people).

Omission training (lower left) occurs *when the omission of the response produces reinforcement*. Therefore, producing the response also leads to a lack of reinforcement. For example, "If you make one more snotty remark, you will get no dessert!" Omitting the snotty remark is reinforced with dessert; producing the response leads to a lack of dessert. (This procedure is occasionally called *negative punishment* to indicate that the response is punished by the absence of an event that would have otherwise occurred.)